3.0 REQUIRED LAB EQUIPMENT & PARTS:

- DC Power Supply (PS), Function Generator (FG) and Oscilloscope.
- ELE202 Lab Kit and ELE302 Lab Kit: various components, breadboard, wires and jumpers.

4.0 PRE-LAB ASSIGNMENT (3 marks with 1 mark for each step):

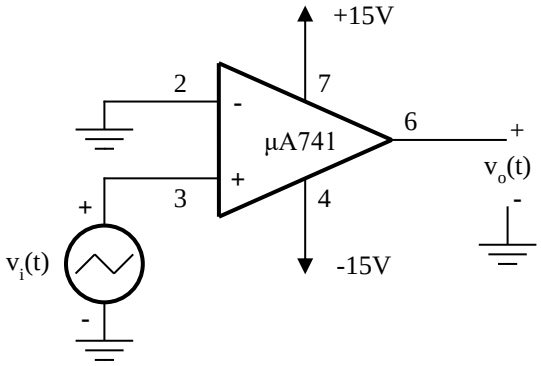
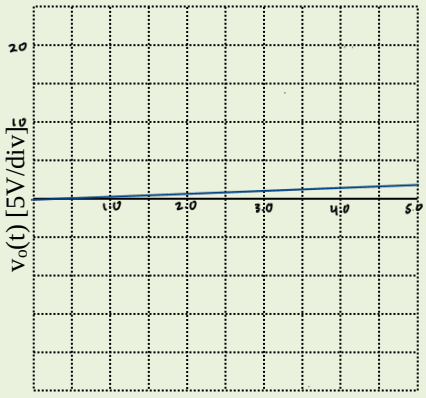


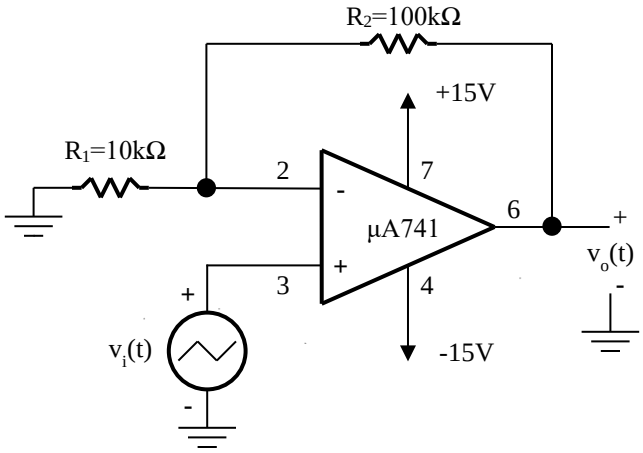
Figure 2.0: Op-Amp without Feedback

(a) Step 1: The circuit in **Figure 2.0** shows the pin identification for the μA741 Op-Amp and the connections of the two dc-power supplies that are required for its operation. In particular, pin (2) is connected to the common ground (COM), while a triangular wave, voltage $v_i(t)$, is applied to pin (3), use the **Graph 1.0a** to show how the resulting voltage-transfer characteristics, [$v_o(t)$ vs $v_i(t)$], would look like, given that the output saturation voltages of the Op-Amp are ±15V and the equivalent-circuit model of the Op-Amp is shown in **Figure 1.0b**.

Pre-Lab workspace



Graph 1.0a



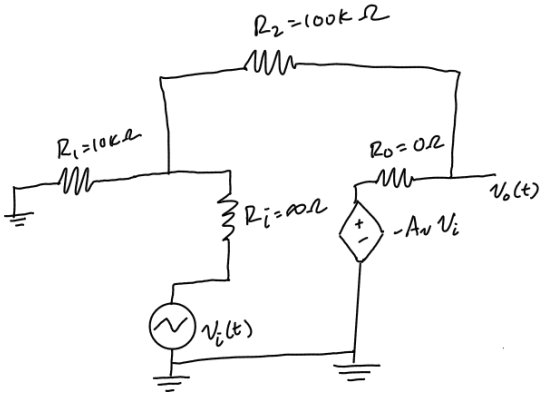


Figure 3.0: Non-Inverting Configuration

(b) Step 2: Assume that the circuit shown in **Figure 3.0** is working properly:

i) Use the equivalent-circuit model of the ideal Op-Amp to find the voltage gain, $A_v = v_o(t) / v_i(t)$, of the amplifier circuit.

Pre-Lab workspace (show your analysis here)

$$V_i(t) = V_o(t) \frac{R_1}{R_1 + R_2}$$

$$A_v = \frac{V_o(t)}{V_i(t)} = \frac{R_1 + R_2}{R_1} = \frac{10k + 100k}{10k} = 11$$

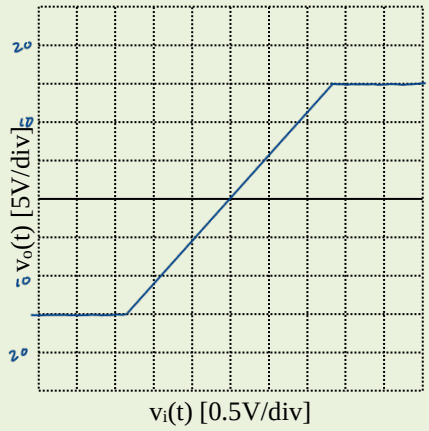
ii) Use the equivalent-circuit model of the μA741 shown in **Figure 1.0b** instead to find A_v.

Pre-Lab workspace (show your analysis here)

$A_V = \frac{V_o(t)}{V_{id}(t)} = 0.5 \times 10^6 = 500\,000$

iii) Use **Graph 1.0b** to plot the voltage transfer characteristics, given that the output saturation voltages of the Op-Amp are ±15V.

Pre-Lab workspace



Graph 1.0b

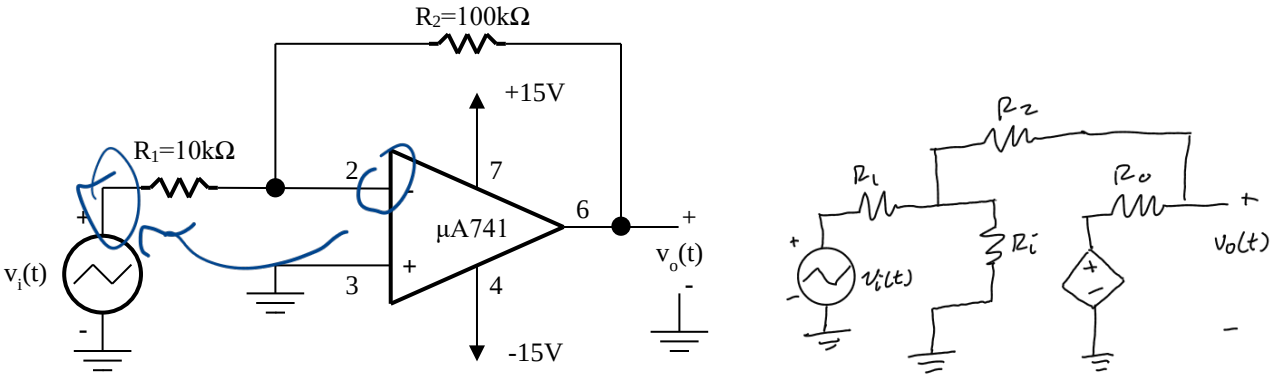


Figure 4.0: Inverting Configuration

(c) Step 3: Assume that the circuit in **Figure 4.0** is working properly.

i) Use the equivalent-circuit model of the ideal Op-Amp instead to determine the values for A_v.

Pre-Lab workspace (show your analysis here)

$$\frac{v_i(t) - 0}{R_1} = \frac{0 - v_o(t)}{R_2}$$

$$A_v = \frac{v_o(t)}{v_i(t)} = -\frac{R_2}{R_1} = -\frac{100k}{10k} = -10$$

ii) What is the value of the voltage at node (2)?

Pre-Lab workspace (show your analysis here)

0V as we know $V_+ = V_-$ and since V_+ is connected directly to the ground, it has 0V and therefore V_- is also virtually grounded at 0V.

5.0 IN-LAB IMPEMENTATION & MEASUREMENTS (5 marks in total):

(a) Step 1: Connect the circuit as shown in **Figure 2.0**. Connect pin (2) of the Op-Amp to common ground. Use a function generator to provide the input voltage, $v_i(t)$, by connecting the function generator output to pin (3) of the Op-Amp. Connect Channel (2) of the oscilloscope to display the output voltage $v_o(t)$, which is between pin (6) and the common ground. Connect Channel (1) of the oscilloscope to display the input voltage $v_i(t)$ provided by the function generator.

Press "AutoScale" to capture Channel (1) and Channel (2) signals on the oscilloscope display.

Adjust this initial display by setting the oscilloscope "Vertical Controls" as follows:

- Channel (1): Coupling => DC, V/div => 2V.
- Channel (2): Coupling => DC, V/div => 5V.

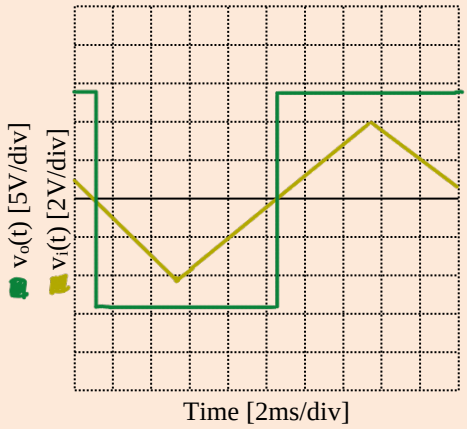
Setting the oscilloscope "Trigger" and "Horizontal Controls" as follows:

- Time base: Trigger Source -> Channel (1), Trigger Type => Edge, Time/div => 2ms.

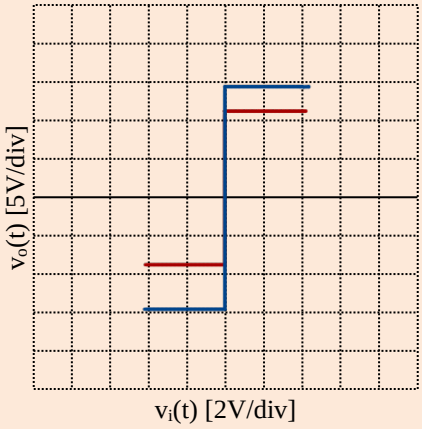
(b) Step 2: Set the controls of the function generator to provide triangular input signal $v_i(t)$ of 8V (peak-to-peak) at a frequency of 50Hz. Use **Graph 2.0** to plot $v_i(t)$ & $v_o(t)$ as functions of time.

(c) Step 3: Change the oscilloscope setting to the XY-display mode. The oscilloscope are now displaying: [$v_o(t)$ vs $v_i(t)$], which is called the voltage-transfer characteristics of the Op-Amp Circuit. Use **Graph 3.0** to plot this voltage-transfer [$v_o(t)$ vs $v_i(t)$] characteristics of the Op-Amp circuit.

(d) Step 4: Change the values of the positive dc-supply voltage to 12V and the negative dc-supply voltage to 9V. Use **Graph 3.0** to plot the resulting voltage-transfer characteristics, [$v_o(t)$ vs $v_i(t)$].



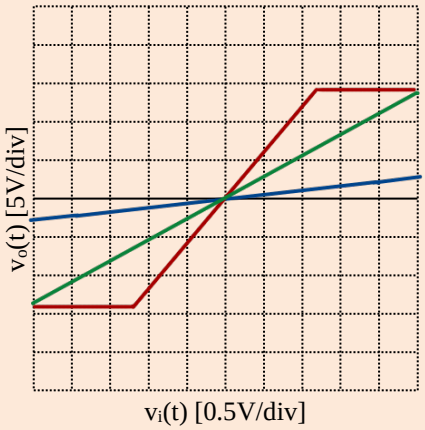
Graph 2.0 (0.5 marks)



Graph 3.0 (0.5 marks)

(e) Step 5: Change the values of both the positive & negative dc-supply voltages to +15V and -15V, respective, and modify your circuit as shown in **Figure 3.0**. Negative feedback is now said to be applied around the Op-Amp, with feedback factor $\beta = R_1/(R_1 + R_2)$, and the circuit is said to operate as a non-inverting amplifier. Plot the resulting voltage-transfer characteristics on **Graph 4.0**. Record feedback factor β in **Table 1.0**. Use your graph to record the Dynamic Range and Voltage Gain also in **Table 1.0**. (The **Dynamic Range** is the linear region of the voltage-transfer characteristics**.)**

(f) Step 6: This concludes the first 3-hr lab session. Please demonstrate Step 1 to Step 5 to your TA and submit your answers to the prelab assignments and the results collected from Step 1 to Step 5 to your TA at the end of the lab session. (1 mark)



$\Delta x = 2.700\ V$
$\Delta y = 28.875\ V$
$\Delta x = 5.05\ V$
$\Delta y = 28.9375\ V$
$\Delta x = 8.6875\ V$
$\Delta y = 8.325\ V$

Graph 4.0 (0.5 marks)

Table 1.0 (0.5 marks)

Circuit Conditions	Feedback Factor β	Valid Input Range (Dynamic Range) for $v_i(t)$	Voltage Gain $[v_o(t)/v_i(t)]$
Negative feedback [R_1=10kΩ & R_2=100kΩ]	0.091	-1.2 , 1.2	10.69
Negative feedback [R_1= 10kΩ & R_2=47kΩ]	0.175	-2.525 , 2.525	5.73
Negative feedback [R_1=inf & R_2=0]	1.0	-4.225 , 4.225	1.04

$\frac{\Delta y}{\Delta x} = \frac{28.875}{2.700} = 10.69$

$\frac{28.9375}{5.05} = 5.73$

$\frac{8.6875}{8.325} = 1.04$

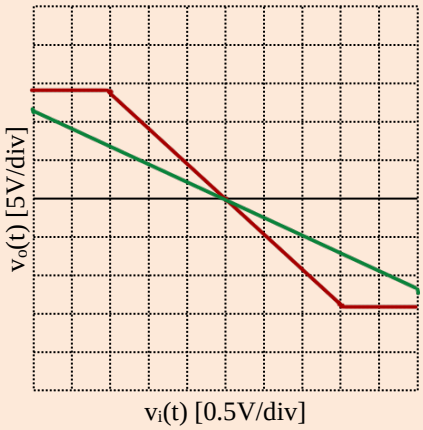
(g) Step 7: Replace R_2 with a 47kΩ resistor on the **Figure 3.0** circuit. Use **Graph 4.0** to plot the resulting XY-mode voltage transfer curve. Use your plot to fill-in **Table 1.0**.

(h) Step 8: By replacing the resistor R_1 with an open-circuit and the resistor R_2 with a short-circuit, your circuit is now said to operate as a voltage follower. Use **Graph 4.0** to plot the resulting voltage-transfer curve, and fill-in the blanks in **Table 1.0**.

(i) Step 9: Modify your circuit as shown in **Figure 4.0**. The circuit is now said to operate as an inverting amplifier. Use **Graph 5.0** to plot the resulting voltage-transfer characteristics and fill-in the blanks in **Table 2.0**.

(j) Step 10: Replace R_2 with a 47kΩ resistor. Use **Graph 5.0** to plot the resulting voltage transfer curve. Use your plot to fill-in the blanks in **Table 2.0**.

(k) Step 11: Demonstrate Step 7 to Step 10 to your TA. (1 mark)



$\Delta x = 2.88\,V$
$\Delta y = -28.625\,V$
$\Delta x = 5.88\,V$
$\Delta y = -28.625\,V$

Graph 5.0 (0.5 marks)

Table 2.0 (0.5 marks)

Circuit Conditions	Feedback Factor β	Valid Input Range (Dynamic Range) for $v_i(t)$	Voltage Gain $[v_o(t)/v_i(t)]$
Negative feedback [R_1=10kΩ & R_2=100kΩ]	0.091	-1.49, 1.39	-9.94
Negative feedback [R_1= 10kΩ & R_2=47kΩ]	0.175	-3.03, 2.85	-4.868

$\frac{-28.625}{2.88} = -9.94$

$\frac{-28.625}{5.88} = -4.868$

6.0 POST-LAB QUESTIONS (2 marks in total, 0.5 marks for each question):

(1) By examining your plots on **Graph 3.0**, answer the following:
   a) What are the effects of the dc-supply voltages on the voltage-transfer characteristics?
   b) Would the solo Op-amp qualify as a practical amplifier circuit? Comment on your answer.

a) The dc-supply voltages set the limit for the maximum $v_o(t)$ output. So when it was set to $\pm 15V$, the $V_o(t)$ had a minimum of about $-15V$ and a maximum of about $15V$. The same goes for when it is set to $-9V$ & $12V$.

b) Only if you are using very delicate circuits as since $A = 0.5 \times 10^6$, the linear input range is only about $\pm 30\mu V$ so any realistic input, noise or offset would drive it straight to saturation unless you are dealing with a very delicate circuit that uses very very small amounts of current. Otherwise, it would not be a practical amplifier.

(2) By considering your plots on **Graphs 3.0, 4.0 & 5.0**, answer the following:

a) What does the application of negative feedback (around the Op-amp) have on the amplifier-circuit behavior?

b) Does the application of negative feedback have any effect on the output saturation voltages? Comment on your answer.

a) Negative feedback changes the amplification from 0.5×10^6 to about 11 for not inverting and -10 for inverting. Because of this, the linear output increases from $\pm 30\mu V$ to about $\pm 1.36 V$ for non-inverting and $\pm 1.5 V$ for inverting.

b) No as the saturation voltages stayed the same at about $\pm 15 V$. The only thing that changes is the amount of input required to reach output saturation aka linear range.

(3) Thevenin Equivalent Input Resistance:

a) Calculate the Thevenin equivalent resistance seen by $v_i(t)$ for the non-inverting configuration shown in **Figure 3.0**.

b) Calculate the Thevenin equivalent resistance seen by $v_i(t)$ for the inverting configuration shown in **Figure 4.0**.

c) Which configuration has higher Thevenin equivalent resistance as seen by $v_i(t)$?

a) $i_i = \frac{V_d}{R_{in}} = \frac{V_i/1+A\beta}{R_{in}} = \frac{V_i}{R_{in}(1+A\beta)} = \frac{V_i}{R_{Th}}$

$R_{Th} = R_{in}(1+A\beta) = (2\times10^6\,\Omega)(45456) = \boxed{9.1\times10^{10}\,\Omega}$

b) since pin 3 goes straight to ground and $V_d = \frac{V_o}{A} = \frac{15}{0.5\times10^6} \approx 0$, pin 2 is virtually grounded, therefore R_{in} is only R_1 which is $10k\Omega$.

c) The non-inverting configuration has a higher R_{Th} as $9\times10^{10} > 10\times10^6$.

(4) Thevenin Equivalent Output Resistance:

a) Calculate the Thevenin equivalent resistance seen by $v_o(t)$ for the non-inverting configuration shown in Figure 3.0.

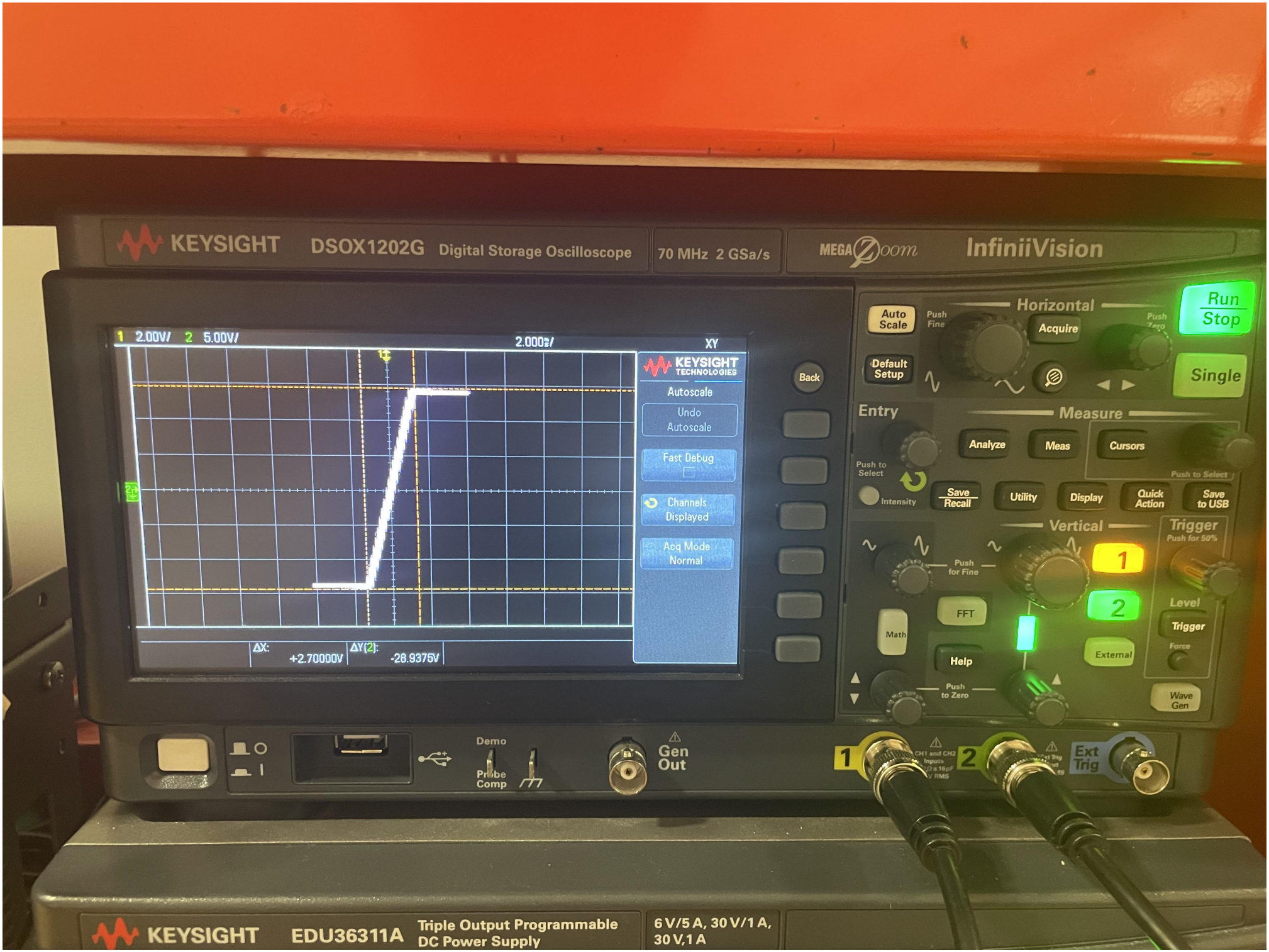
b) Calculate the Thevenin equivalent resistance seen by $v_o(t)$ for the inverting configuration shown in Figure 4.0.

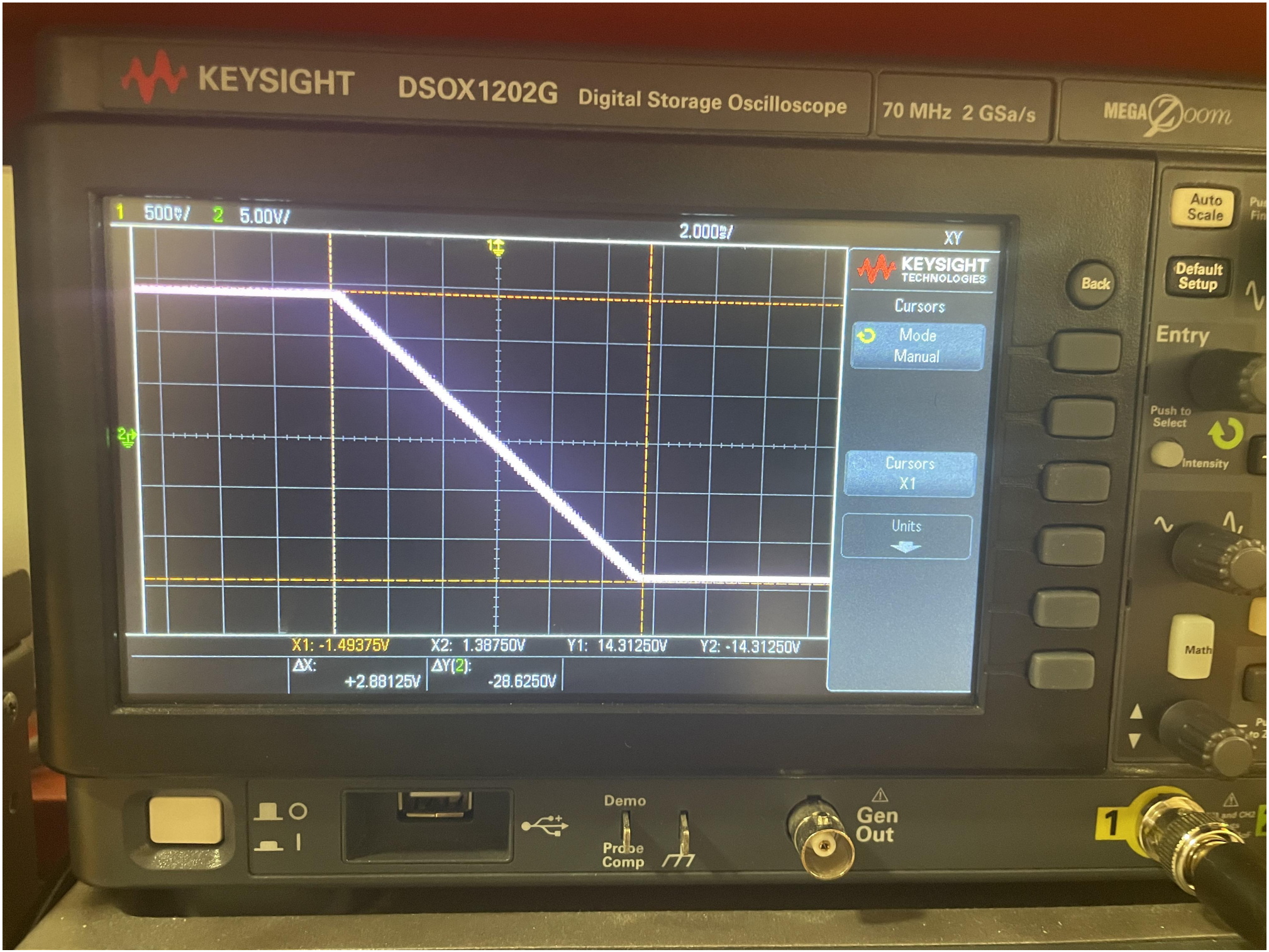
c) How does the two equivalent resistances compare?

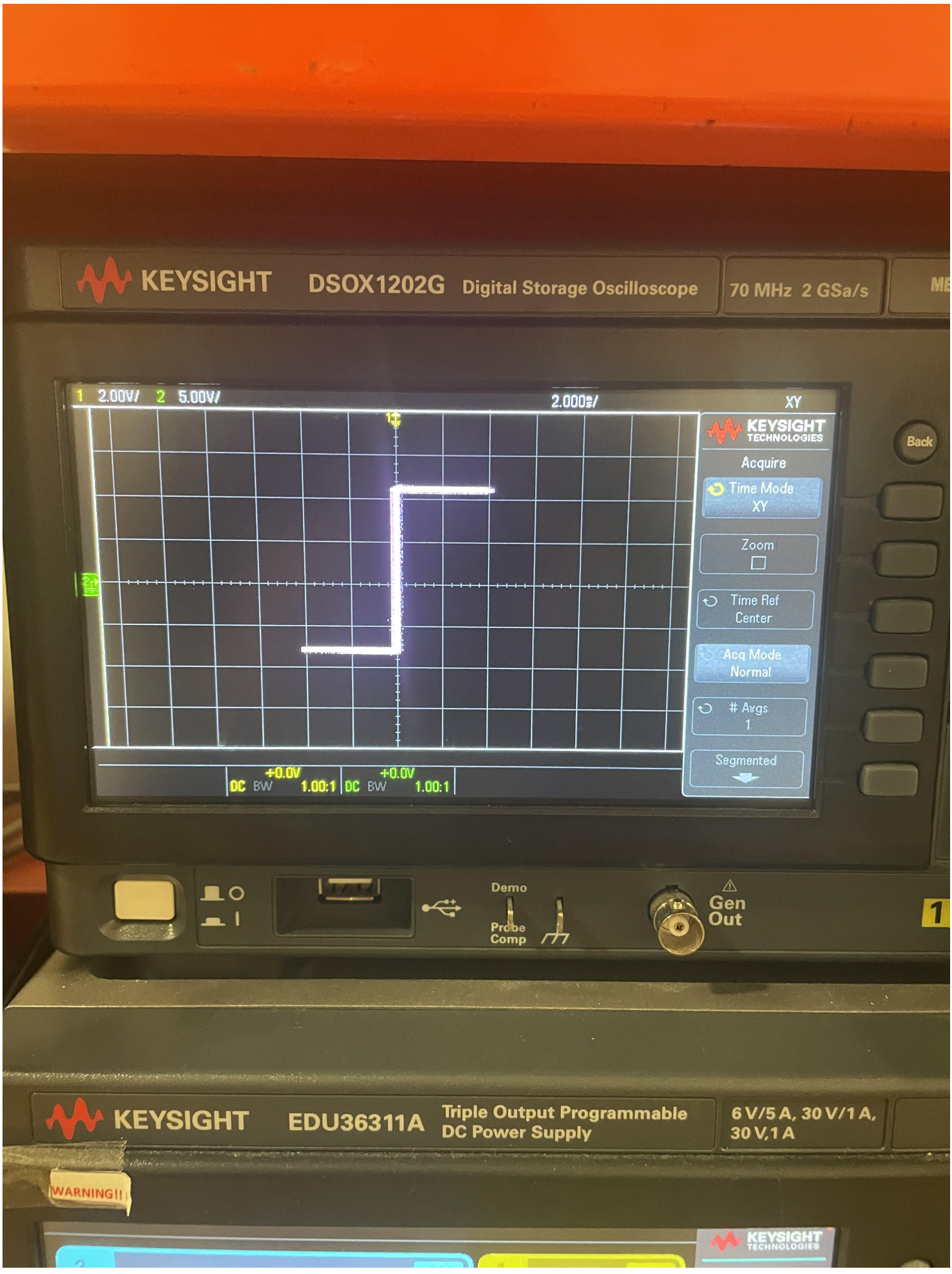
a) $R_{Th} = \frac{R_o}{1+A\beta} = \frac{75}{45456} = 1.65\,m\Omega$

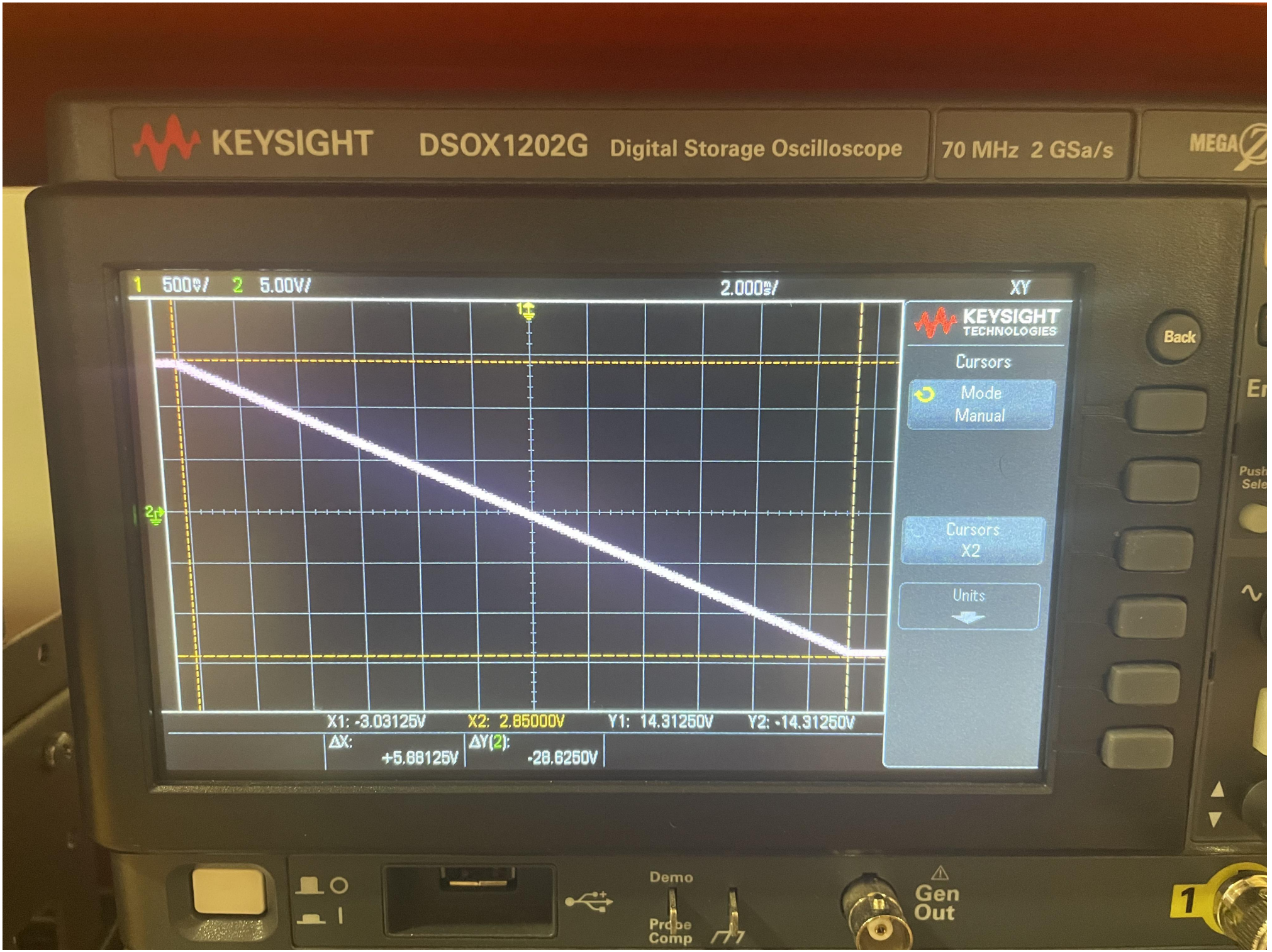
b) $R_{Th} = \frac{R_o}{1+A\beta} = \frac{75}{45456} = 1.65\,m\Omega$

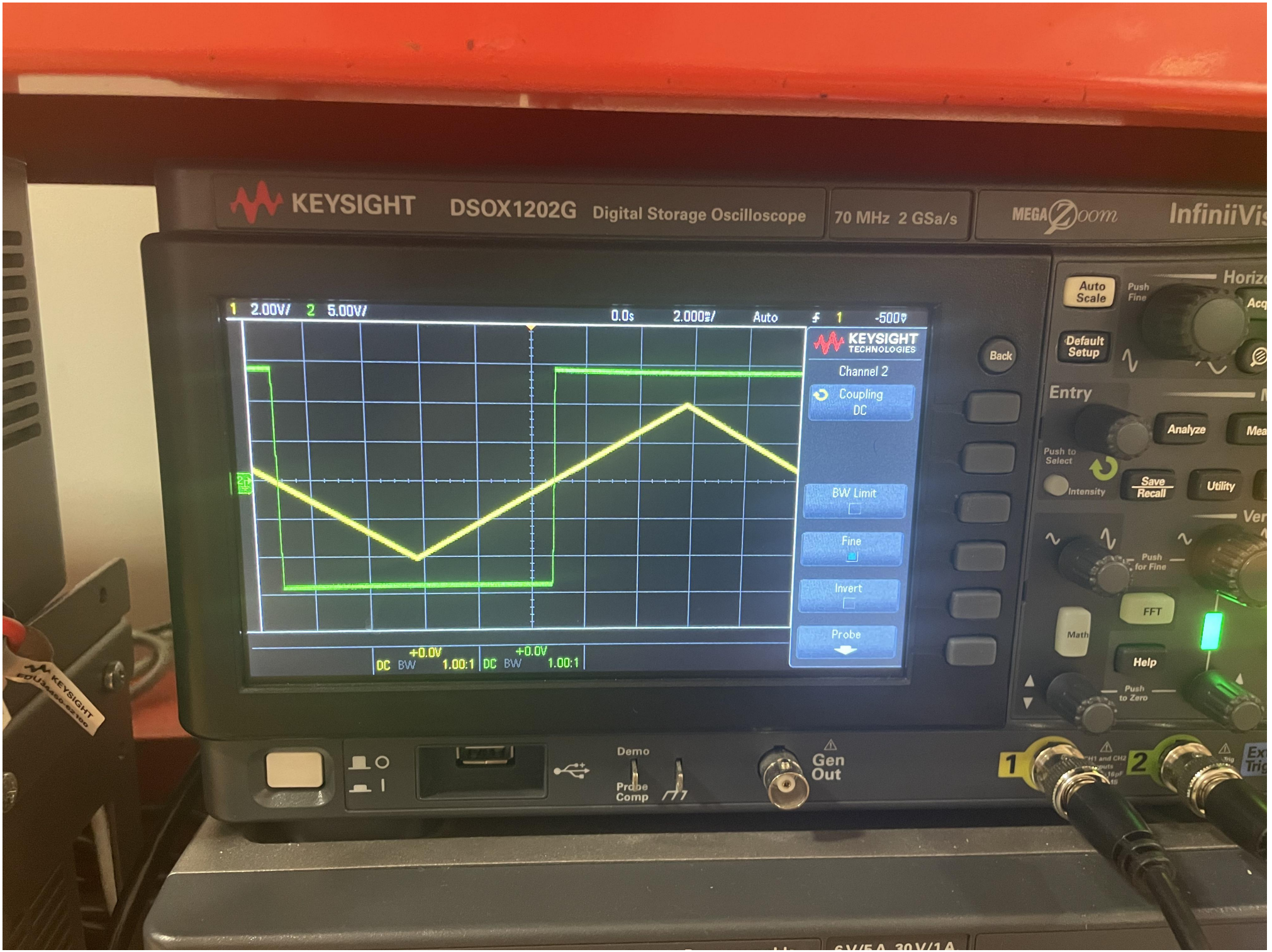
c) The two resistances are the same.

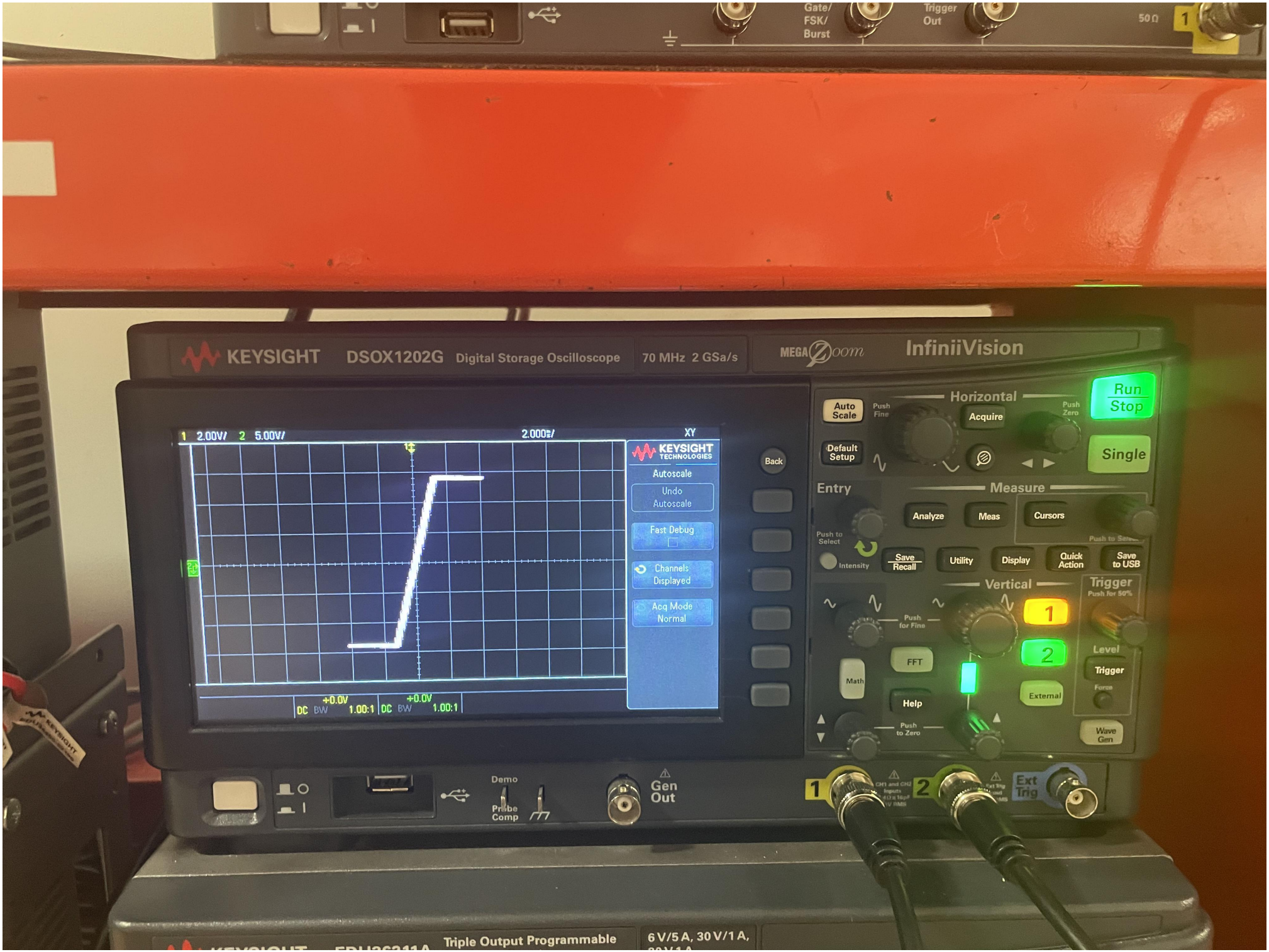

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
MEGAZoom
InfiniiVision
1 2.00V/ 2 5.00V/
2.000s/
XY
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Normal
ΔX: +2.70000V
ΔY(2): -28.9375V
Back
Auto Scale
Push Fine
Horizontal
Acquire
Run Stop
Default Setup
Single
Entry
Measure
Analyze
Meas
Cursors
Push to Select
Intensity
Save Recall
Utility
Display
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Save to USB
Vertical
Trigger
Push for 50%
Push for Fine
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2
FFT
Math
Help
External
Level
Trigger
Force
Push to Zero
Wave Gen
Demo
Probe Comp
Gen Out
Ext Trig
KEYSIGHT
EDU36311A
Triple Output Programmable DC Power Supply
6 V/5 A, 30 V/1 A, 30 V,1 A


KEYSIGHT
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MEGAZoom
1 500mV/ 2 5.00V/
2.000ns/
XY
KEYSIGHT TECHNOLOGIES
Cursors
Mode
Manual
Cursors
X1
Units
X1: -1.49375V
X2: 1.38750V
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Y2: -14.31250V
ΔX:
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ΔY(2):
-28.6250V
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Auto Scale
Default Setup
Entry
Push to Select
Intensity
Math
Demo
Probe Comp
Gen Out


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 2.00V/ 2 5.00V/
2.000s/
XY
KEYSIGHT TECHNOLOGIES
Acquire
Time Mode XY
Zoom
Time Ref Center
Acq Mode Normal
Avgs 1
Segmented
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+0.0V
DC BW 1.00:1
Demo
Probe Comp
Gen Out
KEYSIGHT
EDU36311A
Triple Output Programmable DC Power Supply
6 V/5 A, 30 V/1 A, 30 V,1 A
WARNING!!
KEYSIGHT TECHNOLOGIES


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
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KEYSIGHT TECHNOLOGIES
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Manual
Cursors
X2
Units
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X2: 2.85000V
Y1: 14.31250V
Y2: -14.31250V
ΔX:
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ΔY(2):
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Demo
Probe
Comp
Gen
Out


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
MEGAZoom
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Channel 2
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BW Limit
Fine
Invert
Probe
Back
+0.0V
DC BW 1.00:1
+0.0V
DC BW 1.00:1
Auto Scale
Default Setup
Push Fine
Entry
Analyze
Push to Select
Intensity
Save Recall
Utility
Push for Fine
FFT
Math
Help
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Probe Comp
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Gate/
FSK/
Burst
Trigger
Out
50Ω
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Quick Action
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DC BW +0.0V 1.00:1
Demo
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Triple Output Programmable
6 V/5 A, 30 V/1 A,